# Pandas Tutorial 2: Introduction and Basic Concepts

This chapter introduces Pandas in a clear and beginner friendly way. After completing this tutorial, students will understand what Pandas is, why it is used in Data Science, and how to start working with it.

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# 1. **Introduction to Pandas**

Pandas is a powerful Python library used for data analysis and data manipulation. It is mainly used in Data Science, Machine Learning, and Exploratory Data Analysis.

Pandas helps us work with structured data such as tables, CSV files, Excel files, and databases.

In simple words, Pandas allows us to read data, clean data, analyze data, and prepare data for further processing.

Pandas works very well with other libraries such as NumPy, Matplotlib, and Scikit learn.

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# 2. **Why We Use Pandas in Data Science**

In real world projects, data is usually stored in rows and columns. Working with such data manually is very difficult.

Pandas makes this process simple and fast. It provides built in functions to:

- Read data from files.
- Select specific rows and columns.
- Filter data based on conditions.
- Handle missing values.
- Perform calculations.
- Summarize and analyze datasets.

Because of these features, Pandas is one of the most important libraries for Exploratory Data Analysis.

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# 3. **Installing Pandas**

If you are using a Conda environment, activate your environment first:

```
conda activate eda_env
```

Then install Pandas:

```
conda install pandas
```

Or using pip:

```
pip install pandas
```

After installation, you can import it in Python.

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# 4. **Importing Pandas**

In Python, we usually import Pandas using the alias pd.

```
import pandas as pd
```

The name pd is a standard convention used in almost all Data Science projects.

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# 5. **Core Data Structures in Pandas**

Pandas has two main data structures.

Series

A Series is a one dimensional labeled array. It is similar to a single column in a table.

DataFrame

A DataFrame is a two dimensional table with rows and columns. It is the most important data structure in Pandas.

Most of the time in Data Science, we work with DataFrames.

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# 6. **Creating a Pandas Series**

We can create a Series from a list.

```
import pandas as pd

numbers = [10, 20, 30, 40]
series = pd.Series(numbers)
```

```
print(series)
```

In this example:

- numbers is a Python list.
- pd.Series converts it into a Pandas Series.
- Pandas automatically assigns index values starting from 0.

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# 7. **Creating a Pandas DataFrame**

A DataFrame can be created using a dictionary.

```
import pandas as pd

data = {
    "Name": ["Ali", "Sara", "Ahmed"],
    "Age": [22, 21, 23],
    "Marks": [85, 90, 88]
}

df = pd.DataFrame(data)

print(df)
```

In this example:

- Each key in the dictionary becomes a column name.
- Each list becomes the column values.
- Rows are automatically created.

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# 8. **Viewing Basic Information of DataFrame**

After creating a DataFrame, we often check its basic information.

Head Function

```
df.head()
```

This shows the first 5 rows of the dataset.

Tail Function

```
df.tail()
```

This shows the last 5 rows.

Shape Attribute

```
df.shape
```

This returns the number of rows and columns.

Columns Attribute

```
df.columns
```

This shows the column names.

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# 9. **Selecting Columns from DataFrame**

To select a single column:

```
df["Name"]
```

To select multiple columns:

```
df[["Name", "Marks"]]
```

Notice that double brackets are used when selecting multiple columns.

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# **10. Basic Summary of Data**

Pandas provides a powerful function to summarize numeric data.

```
df.describe()
```

This function gives:

- Count
- Mean
- Standard deviation
- Minimum value
- Maximum value
- Percentiles

This is very useful in Exploratory Data Analysis.

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# 11. Conclusion of Tutorial 1

In this tutorial, students have learned:

1. What Pandas is and why it is important.
2. How to install and import Pandas.
3. The difference between Series and DataFrame.
4. How to create a Series and DataFrame.
5. How to view and summarize data.

In the next tutorial, we will learn how to read real datasets from CSV and Excel files and perform deeper analysis.

This completes Pandas Tutorial 1.